

SCI999 Project

Cluster-Constrained Mean-Variance Optimization

Introduction

Mean-variance optimization [1] is a standard framework for portfolio construction, but in practice it is highly sensitive to estimation error and can produce concentrated portfolios [5]. As a result, strong in-sample performance does not always translate into strong out-of-sample results.

This project examines whether cluster-based diversification constraints can improve the out-of-sample performance of a standard mean-variance portfolio. The approach is to first group similar stocks using unsupervised clustering and then restrict portfolio exposure to any one cluster during optimization. In sample, the unconstrained mean-variance portfolio should perform at least as well as the constrained version because it has greater flexibility. Out of sample, however, the constrained portfolio may perform better if the added diversification reduces concentration risk and sensitivity to estimation error.

1 Data and Universe

The analysis uses daily adjusted price data for S&P 500 constituents. Price histories were loaded from local CSV files derived from Yahoo Finance, while sector classifications were taken from a Bloomberg export dated April 2026. Sector coverage is organized using the Global Industry Classification Standard (GICS), which is the sector taxonomy used throughout the paper. The available price window used in the backtest ends on 13 February 2026. To construct a stable and liquid universe, where liquidity is measured by median daily dollar volume, the following filters are applied:

1. require at least 20 years of available history, removing 101 stocks;
2. remove stocks with any missing-price gap longer than 5 consecutive trading days, removing 1 stock and leaving 395 eligible stocks;
3. forward-fill the remaining small gaps;
4. check for duplicate share classes; none were removed;
5. select the top 3 stocks by median daily dollar volume over a three-month ranking window ending February 2026 from each of the 11 GICS sectors.

All returns used in this study are simple arithmetic returns,

$$r_t = \frac{P_t}{P_{t-1}} - 1.$$

Prices are then resampled to weekly Friday closes. Weekly sampling provides a good balance between lower noise and a still large enough sample for clustering and portfolio estimation. This produces 1,121 weekly return observations.

The sector-stratified selection rule avoids a tech-dominated universe and yields a diversified, liquid sample. Figure 1 shows the correlation structure at different sampling frequencies; the structure is broadly similar across daily, weekly, monthly, and quarterly horizons, confirming that weekly sampling does not distort the main dependence patterns. Table 1 lists the selected companies by GICS sector. Detailed stock-level descriptive statistics are reported in Appendix A. A limitation of the 20-year history filter is that it introduces survivorship bias, meaning firms that disappeared before the end of the sample are excluded.

However, this mainly affects the level of the results, while the relative comparison remains fairer because all strategies are tested on the same filtered universe.

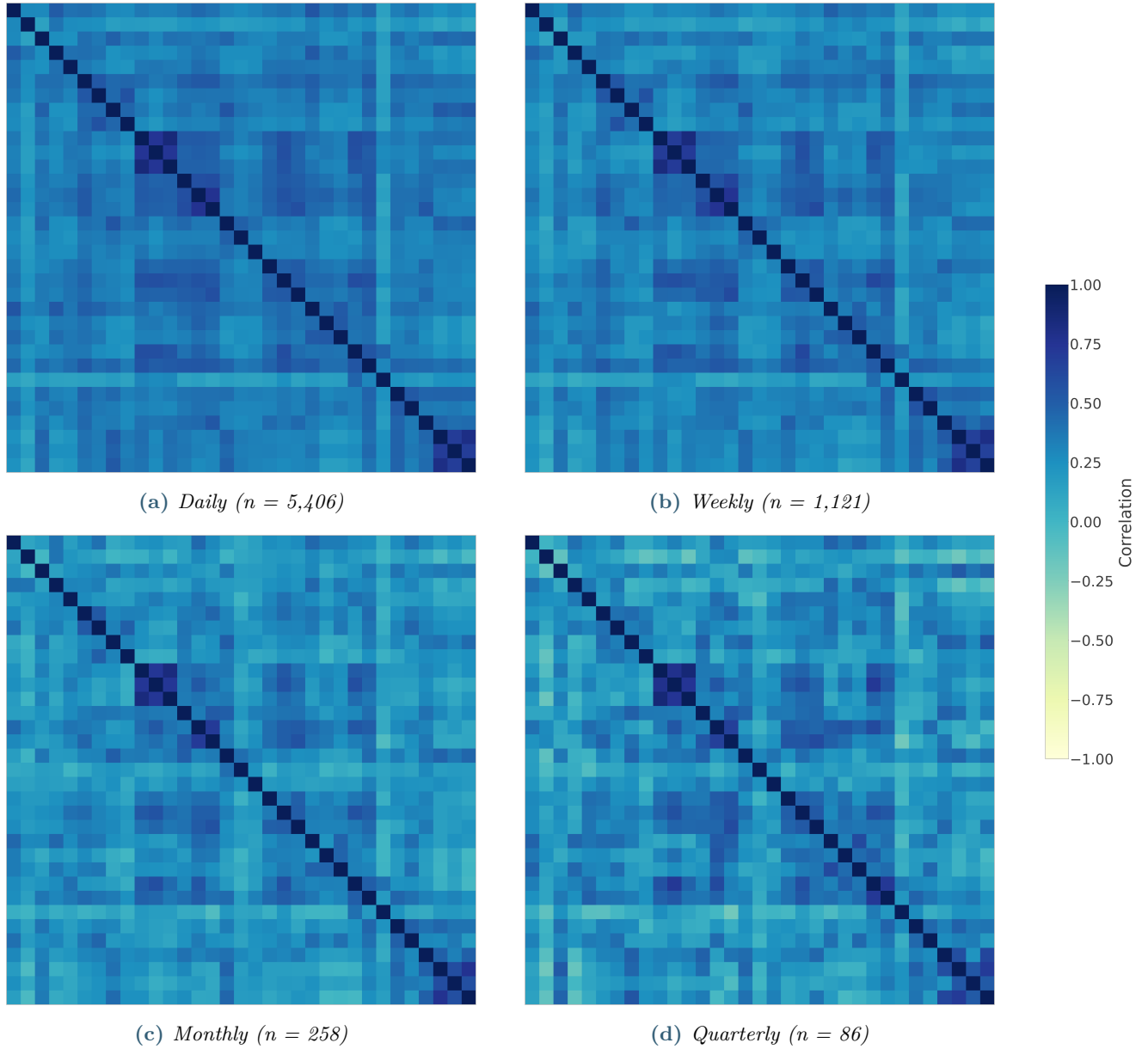


Figure 1: Correlation heatmaps for the selected 33-stock universe at daily, weekly, monthly, and quarterly frequencies. All four panels use the same stock ordering (sorted by GICS sector) and the same color scale (-1 to 1). The broad dependence structure is preserved across frequencies, confirming that weekly sampling does not distort the main patterns.

Table 1: *Universe selection: top 3 liquid companies per GICS sector by median daily dollar volume over a three-month ranking window ending February 2026.*

GICS Sector	Selected Companies
Comm. Services	Alphabet (Class A), Netflix, Verizon
Cons. Discretionary	Amazon, Home Depot, Booking Holdings
Consumer Staples	Walmart, Costco, Procter & Gamble
Energy	ExxonMobil, Chevron, Schlumberger
Financials	JPMorgan Chase, Berkshire Hathaway (Class B), Goldman Sachs
Health Care	Eli Lilly, UnitedHealth Group, Johnson & Johnson
Industrials	Boeing, GE Aerospace, Caterpillar
Info. Technology	Nvidia, Micron Technology, Microsoft
Materials	Freeport-McMoRan, Newmont, Linde
Real Estate	American Tower, Welltower, Equinix
Utilities	NextEra Energy, Duke Energy, Southern Company

2 Clustering

2.1 Clustering Procedure

Clustering is used to define diversification groups, or clusters, for the constrained mean-variance optimizer. At each weekly rebalance date, the 33-stock universe is represented by the previous five years of weekly returns, from which a Pearson correlation matrix is estimated. Correlations are converted into distances using the correlation-distance metric introduced by [4],

$$d_{ij} = \sqrt{\frac{1 - \rho_{ij}}{2}},$$

where ρ_{ij} is the Pearson correlation between stocks i and j , so that highly correlated stocks are close and weakly correlated stocks are farther apart.

Hierarchical agglomerative clustering with Ward linkage [3] is then applied to this distance structure. Ward’s method merges the pair of groups that produces the smallest increase in within-cluster dispersion [12]. The merge cost is

$$\Delta(A, B) = \frac{|A||B|}{|A| + |B|} \|\bar{x}_A - \bar{x}_B\|^2,$$

where $|A|$ and $|B|$ are the sizes of two clusters being merged, and $\bar{x}_A$ and $\bar{x}_B$ are their centroids (the mean return vectors of each cluster). Candidate values of the number of clusters, denoted k , are compared using silhouette quality, Davies-Bouldin separation, and rolling-window stability. The dendrogram is then cut at the selected value of k .

Before presenting the clustering results, the main diagnostics are:

- Silhouette score [6]: compares how close a stock is to its own cluster relative to the nearest alternative cluster; higher values indicate tighter clusters.

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}},$$

where $a(i)$ is the average distance from stock i to other stocks in its own cluster and $b(i)$ is the lowest average distance to stocks in any other cluster.

- Davies-Bouldin index: measures average similarity between each cluster and its closest alternative cluster; lower values indicate better separation.
- Singleton count: counts clusters containing only one stock. Too many singletons make the cluster cap less useful because the constraint starts acting like a stock-level cap.
- Rolling stability: measures whether cluster assignments remain similar through time. I use the Adjusted Rand Index (ARI)¹ and cluster persistence² for this.

2.2 Selecting the Number of Clusters

I evaluate $k = 4$ through $k = 9$ using full-period fit and rolling-window practicality. As shown in Table 2 and Figure 2, silhouette scores improve as k rises, while the Davies-Bouldin index falls, so full-period fit alone would favor larger k . However, higher k also creates narrower and less practical groups.

The main trade-off appears in the rolling diagnostics. At $k = 5$, the partition remains too coarse because it keeps Newmont (NEM), a gold miner, inside the broader commodity-sensitive cluster. Moving to $k = 6$ isolates Newmont, which is economically sensible because gold miners often behave differently from broader commodity-cyclical firms: their returns are more tied to gold-price and defensive dynamics than to general industrial demand. Meanwhile, higher values of k sharply increase the rolling singleton rate from 65.3% at $k = 6$ to 100.0% at $k = 9$. I therefore retain $k = 6$ as the baseline because the paper prioritizes practical, stable clustering over ex-post Sharpe maximization. The sensitivity analysis in Section 4 confirms that higher k can raise Sharpe but at the cost of more degenerate cluster structures.

Table 2: Full-period and rolling diagnostics for cluster selection. Larger silhouette values are preferable, while lower Davies-Bouldin values indicate better-separated clusters. Rolling singleton rate measures how often at least one one-stock cluster appears in rolling five-year windows.

k	Silhouette	Davies-Bouldin	Full-Period Singletons	Rolling Singleton %
4	0.0625	2.368	0	1.4
5	0.0748	2.208	0	38.0
6	0.0739	1.934	1	65.3
7	0.0728	1.833	1	79.3
8	0.0836	1.786	1	89.7
9	0.0862	1.711	1	100.0

¹The Adjusted Rand Index measures similarity between two cluster assignments while adjusting for chance agreement. A value of 1 means identical partitions.

²Higher values indicate more stable cluster memberships from one window to the next.

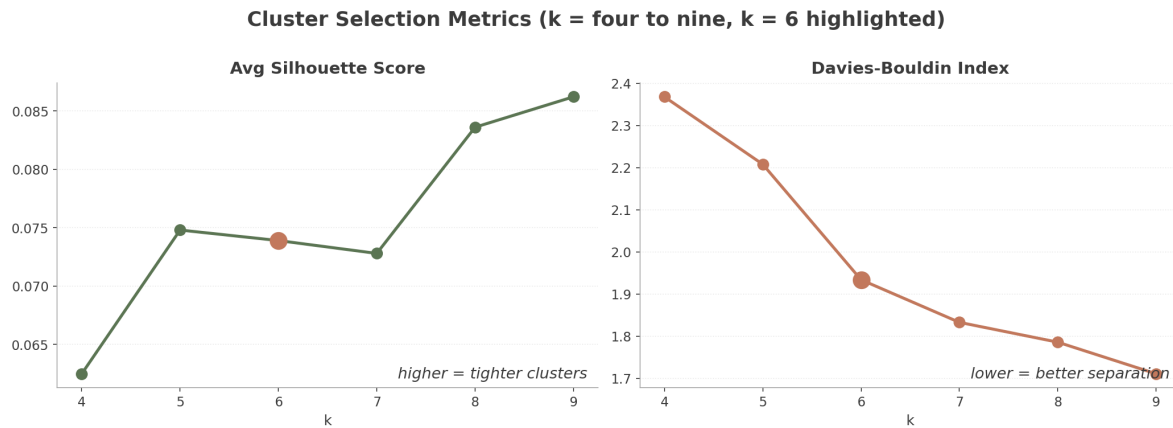


Figure 2: Cluster-selection diagnostics for $k = 4$ through $k = 9$. The figure shows the trade-off between statistical fit and practical cluster structure; the selected baseline is $k = 6$.

2.3 Full-Period Cluster Structure

Having selected $k = 6$, I next examine the full-period clustering. Figure 3 shows the correlation heatmap before, ordered by GICS, and after, ordered by the selected clusters. Ordering by the selected clusters reveals clearer block structure than ordering by GICS alone. Figure 4 shows the Ward dendrogram, and Table 3 reports the six resulting groups: utilities, defensives, commodity cyclicals, gold, growth and technology, and broad cyclicals. These do not match GICS sectors one-for-one, but they are economically intuitive and reflect common return behavior rather than administrative sector labels.

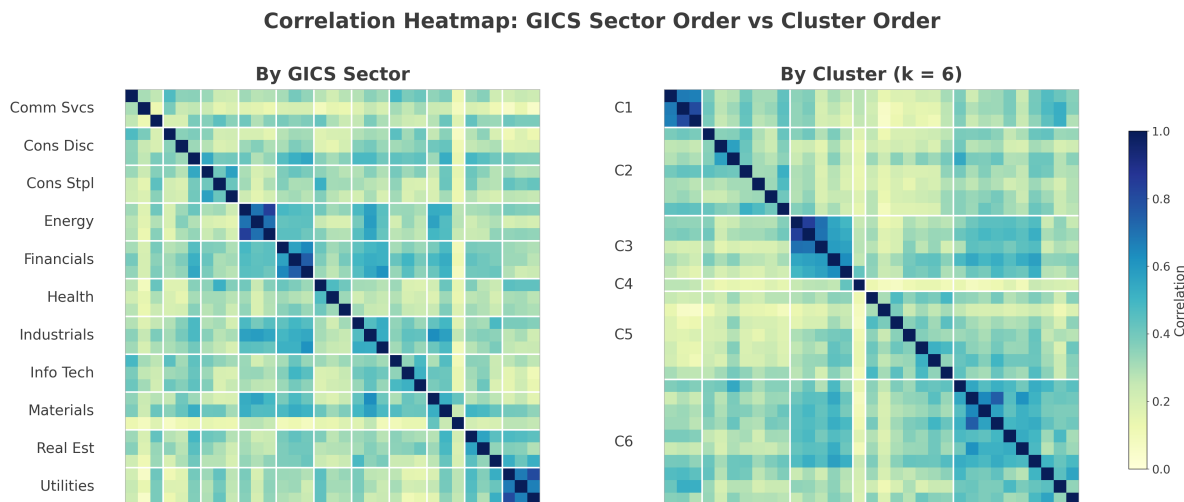


Figure 3: Weekly return correlation heatmaps ordered by GICS sector on the left and by the selected $k = 6$ clusters on the right. The cluster ordering makes the return-based grouping visible rather than relying only on sector labels.

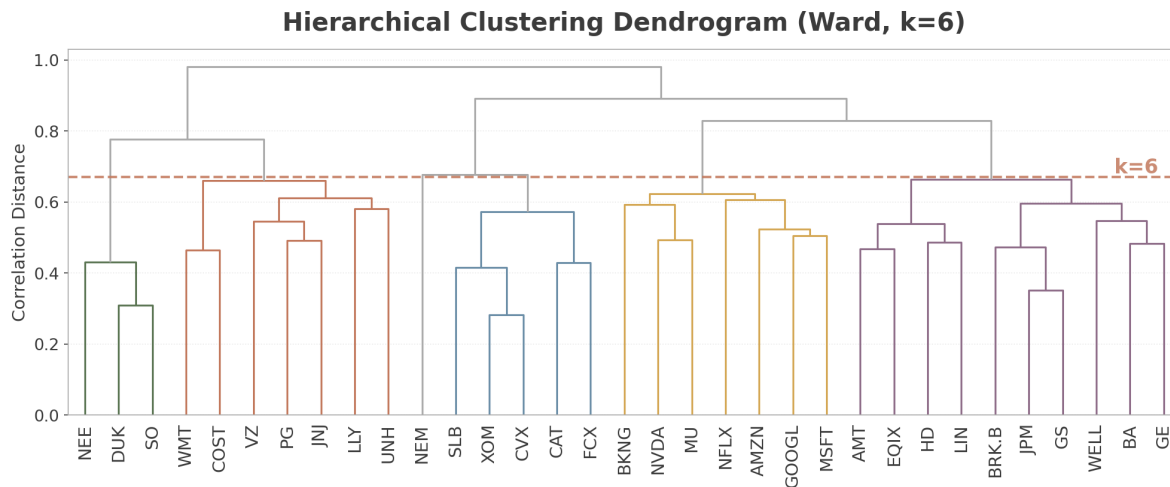


Figure 4: Full-period Ward dendrogram (tree diagram of hierarchical cluster merges) for the 33-stock universe. The selected cut at $k = 6$ balances interpretability with the need to avoid excessive singleton clusters.

Table 3: Full-period cluster composition at $k = 6$. The labels are descriptive, not imposed by GICS; they summarize common return behavior in the clustering output. Several clusters cut across GICS sectors, indicating that the grouping is driven by return co-movement rather than sector labels alone.

Cluster	Description	Size	Constituents
C1	Utilities	3	NextEra Energy, Duke Energy, Southern Company
C2	Defensives	7	Verizon, Walmart, Costco, Procter & Gamble, Eli Lilly, UnitedHealth, Johnson & Johnson
C3	Commodity cyclicals	5	ExxonMobil, Chevron, Schlumberger, Caterpillar, Freeport-McMoRan
C4	Gold	1	Newmont
C5	Growth and technology	7	Alphabet, Netflix, Amazon, Booking Holdings, Nvidia, Micron, Microsoft
C6	Broad cyclicals	10	Home Depot, JPMorgan, Berkshire Hathaway, Goldman Sachs, Boeing, GE Aerospace, Linde, American Tower, Welltower, Equinix

The labels in Table 3 are descriptive names assigned after clustering; they are not imposed by GICS. For example, Newmont forms its own gold cluster, while the defensive cluster combines consumer staples, health care, and Verizon, showing that the grouping is based on similar return behavior rather than formal sector labels.

2.4 Rolling 5-Year Stability

Finally, I test whether the $k = 6$ partition is stable through time. The clustering procedure is repeated in 862 rolling five-year windows. Figure 5 visualizes the resulting memberships. The mean Adjusted Rand Index across consecutive windows is 0.939 and mean cluster persistence is 0.945, indicating a high degree of stability in cluster memberships from one window to the next. This supports using a fixed $k = 6$ interpretation for the portfolio constraints.

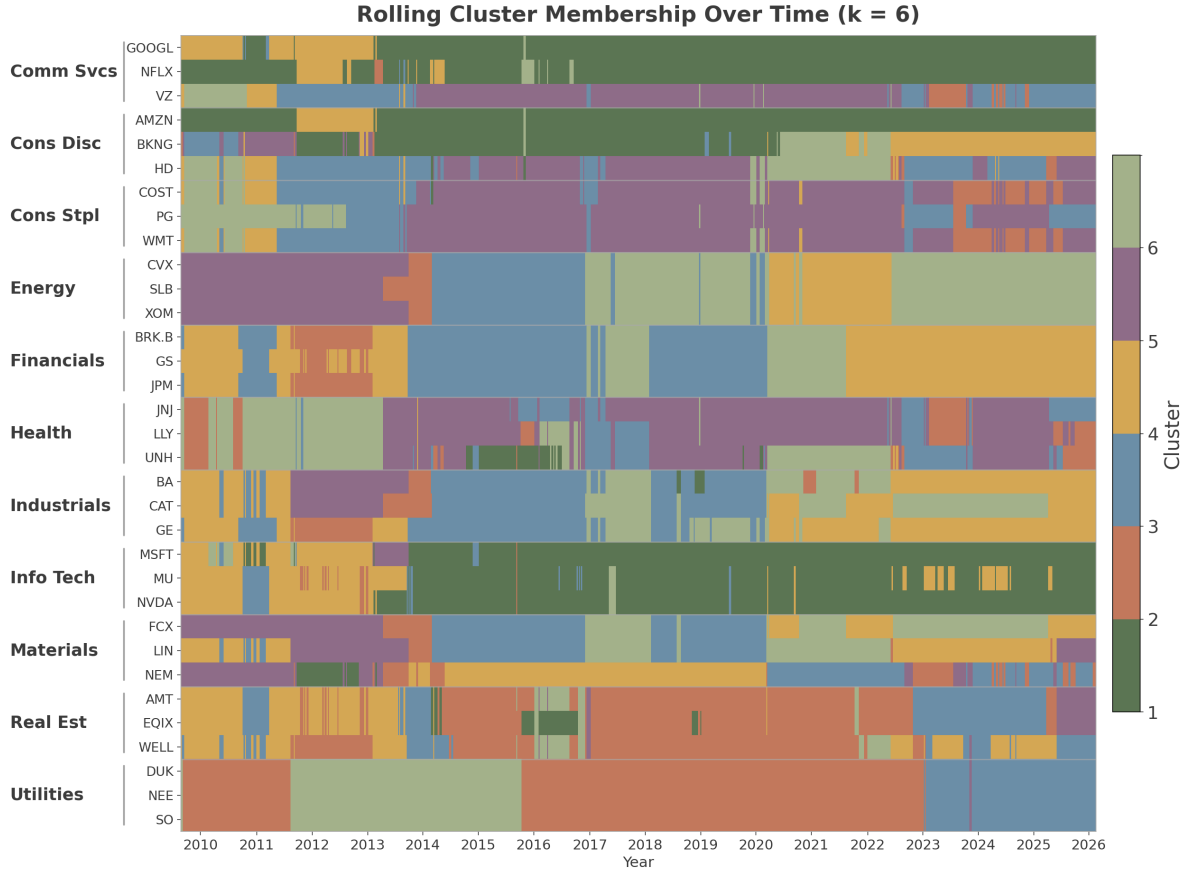


Figure 5: Rolling five-year cluster membership for the selected $k = 6$ specification. Stable color bands indicate that most stocks remain in consistent clusters across adjacent windows.

3 Methodology

This section describes the portfolio construction framework and the weekly walk-forward backtest used to evaluate it. Three main portfolio specifications are compared: cluster-constrained mean-variance optimization (CMVO), unconstrained mean-variance optimization (Plain MV), and the equal-weight $1/N$ benchmark [8]. Section 4 also considers an extension of CMVO that adds a 5% maximum weight per stock to examine whether residual stock-level concentration can be reduced further.

3.1 Portfolio Specifications

Let μ_t and Σ_t denote the expected return vector and covariance matrix estimated from the previous five years of weekly data at rebalance date t . Specifically, μ_t is estimated using an annualized historical return estimator applied to the trailing five-year weekly price window, and Σ_t is the sample covariance matrix of the same returns, annualized to a 52-week year. Let $w_t \in \mathbb{R}^{33}$ denote portfolio weights, and let $\mathcal{C}_1, \dots, \mathcal{C}_6$ denote the clusters from Section 2.

Plain MV uses the same estimation window and optimization framework as CMVO but without cluster constraints. For the maximum-Sharpe version [2], Plain MV solves

$$\max_{w_t} \frac{\mu_t^\top w_t}{\sqrt{w_t^\top \Sigma_t w_t}} \quad \text{s.t.} \quad \mathbf{1}^\top w_t = 1, \quad 0 \leq w_{i,t} \leq \bar{w}.$$

For the minimum-variance version, it solves

$$\min_{w_t} w_t^\top \Sigma_t w_t \quad \text{s.t.} \quad \mathbf{1}^\top w_t = 1, \quad 0 \leq w_{i,t} \leq \bar{w}.$$

For the baseline CMVO and Plain MV specifications, $\bar{w} = 1$, so there is no binding stock-level cap beyond the long-only constraint.

CMVO uses the same objectives, but adds the cluster exposure cap

$$\sum_{i \in \mathcal{C}_c} w_{i,t} \leq 0.30, \quad c = 1, \dots, 6.$$

With six clusters, equal allocation is approximately 16.7% per cluster, so the 30% cap allows the optimizer roughly double the equal share before the constraint binds. The 5% stock-cap extension keeps the cluster cap and sets $\bar{w} = 0.05$; with 33 stocks, this is about 1.65 times the equal-weight allocation of 3.03%, tight enough to prevent extreme positions but loose enough for the optimizer to express views. Neither cap value was optimized; both are round-number constraints chosen to allow moderate deviation from equal allocation while preventing extreme concentration. The equal-weight benchmark is not optimized; it assigns

$$w_{i,t} = \frac{1}{33}, \quad i = 1, \dots, 33.$$

3.2 Inputs and Constraints

At each rebalance date, expected returns and the covariance matrix are estimated from the previous five years of weekly data. The baseline covariance estimator is the sample covariance matrix. All optimized portfolios are fully invested and long-only, meaning no short positions are allowed. Transaction costs and slippage are not modelled; turnover is reported so that the reader can assess implementation cost. In CMVO, the six-cluster structure is re-estimated at each rebalance using Ward hierarchical clustering and then used as the basis for the cluster weight cap. The use of practical weight constraints is motivated by [7], who show that constraints can help stabilize mean-variance portfolios in the presence of estimation error.

3.3 Walk-Forward Design

The backtest is implemented using a weekly walk-forward design that uses only information available at the time of each rebalance. At week t , portfolio weights are estimated from the trailing five-year window of returns and are then held for the following week, with performance evaluated out of sample in week $t + 1$:

$$r_{p,t+1} = w_t^\top r_{t+1}.$$

The estimation window is rolled forward by one week at each step, and the procedure is repeated over the full sample.

The common out-of-sample comparison begins on 18 February 2011 because the optimized strategies require a full five-year estimation window. The common evaluation window ends on 13 February 2026 and contains 783 realized weekly out-of-sample return observations. Figure 6 illustrates the rolling-window structure.

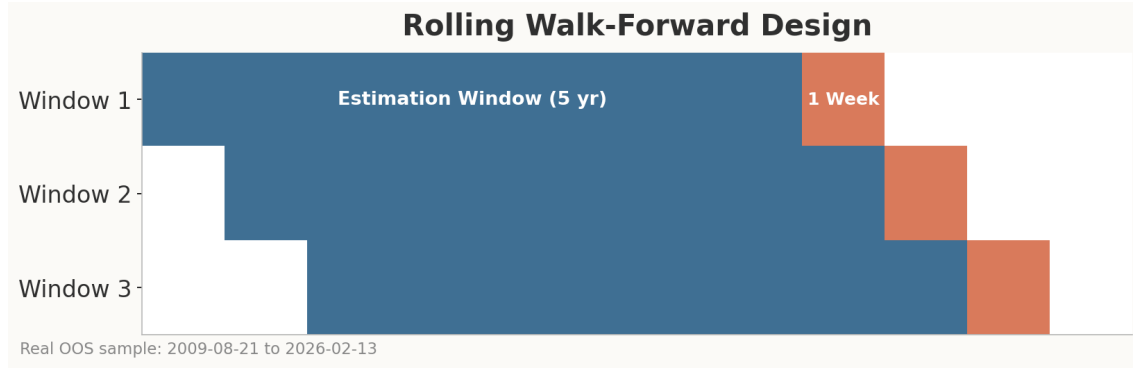


Figure 6: *Rolling walk-forward window design using a five-year estimation window and one-week out-of-sample test period (first 3 of 862 windows shown, not to scale). Each window estimates weights using past data only and evaluates the next realized week.*

3.4 Evaluation Metrics

Out-of-sample performance is evaluated using annualized return, annualized volatility, and the annualized Sharpe ratio. Downside risk is summarized using maximum drawdown and expected shortfall at the 5% level (ES 5%)³. Portfolio concentration is measured using the Herfindahl-Hirschman Index (HHI)⁴, and trading intensity is measured using average turnover⁵. Pairwise Sharpe-ratio differences are tested using the Jobson and Korkie (1981) statistic with the Memmel correction [10, 9].⁶ This test checks whether two Sharpe ratios are statistically different rather than just numerically different.

Table 4 summarizes the main hyper-parameters and modelling choices. These are called hyper-parameters because they are part of the experimental setup and are not directly optimized by the portfolio objective.

Table 4: *Summary of methodology. These hyper-parameters define the experimental setup used for the baseline backtest.*

Hyper-parameter or choice	Value
Estimation window	5 years of weekly returns
Rebalance frequency	Weekly
Number of clusters	$k = 6$
Linkage method	Ward
Cluster weight cap	30% per cluster
Baseline optimization objective	Maximum Sharpe ratio
Covariance estimator	Sample covariance
Universe	33 S&P 500 stocks, 3 per GICS sector
Out-of-sample period	18 Feb 2011 to 13 Feb 2026, 783 weeks
Benchmarks	Plain MV and Equal Weight (1/N)

³Expected shortfall at 5% is the average return in the worst 5% of weeks. It is also called conditional value-at-risk in some texts.

⁴HHI is $\sum_i w_i^2$. Larger values indicate a more concentrated portfolio.

⁵Turnover is the average absolute change between drifted portfolio weights and new target weights at each rebalance, following [8].

⁶The Memmel (2003) correction adjusts the original Jobson-Korkie test statistic to improve its finite-sample properties under non-normal returns.

4 Results

4.1 Main Out-of-Sample Comparison

The equal-weight 1/N portfolio and Plain MV are used as the main benchmarks. Because 1/N does not require a five-year estimation window, it is available earlier than the optimized strategies. To keep the comparison fair, all strategies, including 1/N, are aligned to the first common out-of-sample week, 18 February 2011. The common evaluation window ends on 13 February 2026 and contains 783 realized weekly out-of-sample return observations.

Table 5 shows the out-of-sample results. Under the maximum-Sharpe objective, CMVO delivers similar return and Sharpe performance to Plain MV, but with lower concentration and better downside measures. It does come with higher turnover. With the minimum-variance objective, CMVO again offers modest improvements in concentration and downside risk, though both minimum-variance portfolios remain below the equal-weight benchmark in return and Sharpe terms.

Table 5: *Out-of-sample performance comparison over the common 18 Feb 2011–13 Feb 2026 weekly evaluation window. CMVO has competitive Sharpe performance and lower downside and concentration measures than Plain MV, while the Equal Weight benchmark remains difficult to dominate.*

Strategy	Ann. Ret. %	Ann. Vol. %	Sharpe	Max DD %	ES 5%	HHI	Avg Weekly Turnover %
CMVO	21.6	18.3	1.16	-28.0	-5.54	0.1927	15.1
Plain MV	23.7	20.8	1.13	-31.0	-6.36	0.2650	11.3
Equal Weight (1/N)	18.7	16.2	1.14	-32.2	-5.15	0.0303	2.5
CMVO (Min Vol)	12.7	13.4	0.96	-28.3	-4.23	0.1282	9.6
Plain MV (Min Vol)	12.0	13.0	0.94	-28.4	-4.10	0.1533	4.7

In this table, Max DD is the maximum drawdown (largest peak-to-trough decline). Clustering does not materially improve headline Sharpe, but it does improve concentration and downside risk. CMVO preserves competitive risk-adjusted results while producing a more diversified and more defensible portfolio structure. Specifically, CMVO is less risky than Plain MV in several measures: lower HHI (0.1927 vs. 0.2650), shallower maximum drawdown (-28.0% vs. -31.0%), and smaller expected shortfall (-5.54% vs. -6.36%).

Pairwise Sharpe-ratio differences were tested using the Jobson-Korkie statistic with the Memmel correction [10, 9]. None of the differences are statistically significant at conventional levels (all p-values above 0.64), so the contribution of CMVO should be evaluated through portfolio structure and downside risk rather than headline Sharpe.

4.2 Concentration and Downside Behavior

The key difference between CMVO and Plain MV lies in portfolio concentration. Figures 7 and 8 illustrate this clearly. At a sector level, Plain MV repeatedly concentrates large weights in a small number of sectors. At the stock level, this behavior is even more pronounced: Plain MV frequently allocates very large weights to a few names, with the largest position reaching 68.1%.

By contrast, CMVO produces more distributed allocations. Its cluster exposure is capped at 30%, and its exposures are spread across a broader set of names. This difference is reflected in the concentration metrics and in the downside measures: CMVO reduces HHI from 0.2650 to 0.1927, improves ES(5%) from -6.36% to -5.54%, and reduces maximum drawdown from -31.0% to -28.0%.

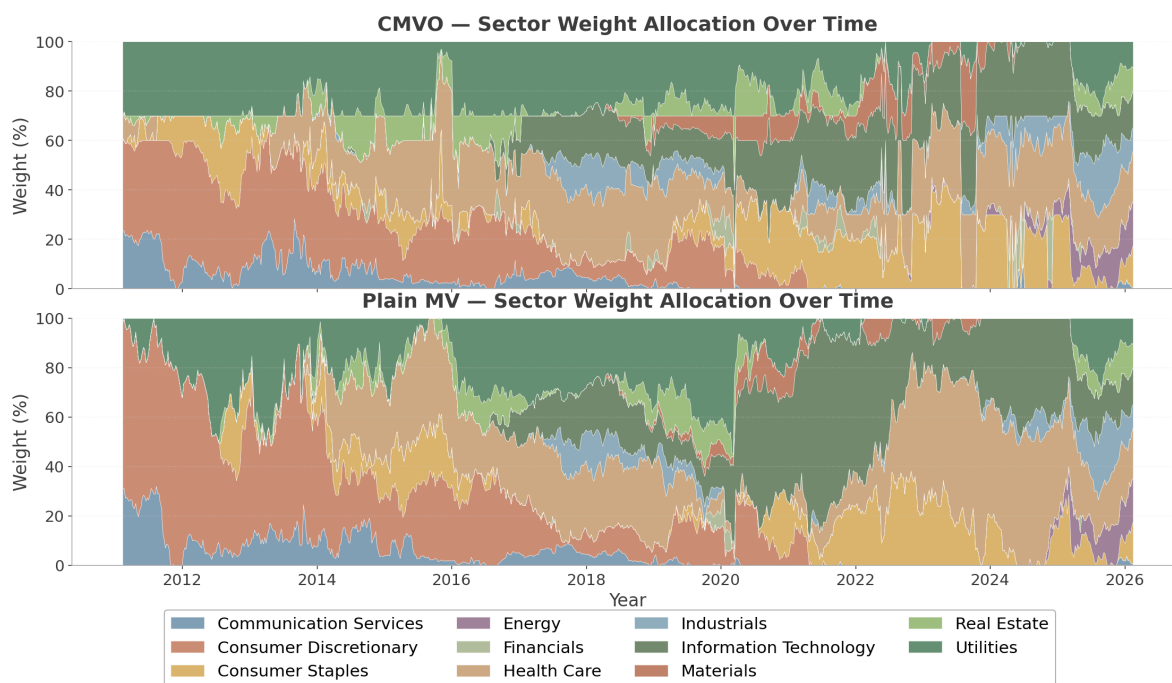


Figure 7: Sector weights over time for CMVO and Plain MV. Plain MV often concentrates in a few sectors, while CMVO spreads exposure more evenly through cluster constraints.

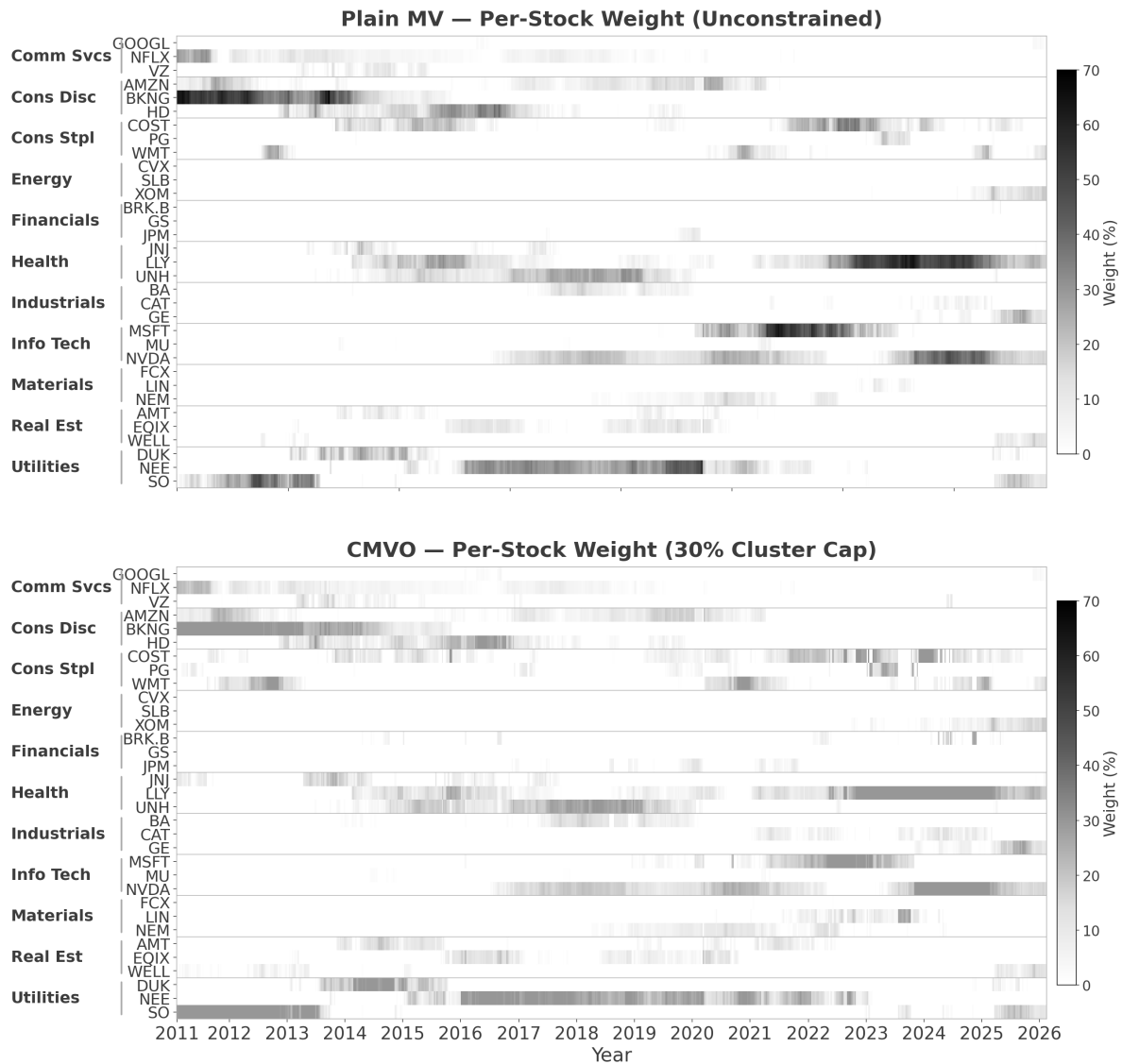


Figure 8: Stock-level weights over time for Plain MV (top) and CMVO (bottom). Each row within a sector band represents one stock; darker shading indicates higher weight. By late 2023 into 2024, Plain MV concentrated almost entirely in Nvidia (NVDA) and Eli Lilly (LLY). CMVO during the same period spread weight across technology, energy, consumer staples, and industrials, though it still exhibits some high single-name weighting within the 30% cluster cap.

4.3 Extension: Adding a 5% Stock Cap

Because the baseline CMVO portfolio still exhibits some stock-level concentration, I test a 5% stock-cap extension to see whether direct position limits can further improve diversification and downside stability. This extension is not the paper’s primary model; it is examined as a robustness variant.

Table 6 shows that the capped portfolio achieves the highest Sharpe ratio in the sample (1.29), while also reducing volatility, drawdown, tail risk, concentration, and turnover relative to baseline CMVO. However, this improvement comes with a cost: annualized return declines from 21.6% to 19.4%. The 5% cap is therefore best interpreted as a stability-enhancing extension rather than a strict improvement along every dimension.

Table 6: 5% stock-cap extension: effect of adding a per-stock weight limit to the baseline CMVO. The cap sharply reduces concentration and improves Sharpe, but annualized return falls.

Strategy	Ann. Ret. %	Ann. Vol. %	Sharpe	Max DD %	ES 5%	HHI	Avg Weekly Turnover %
CMVO	21.6	18.3	1.16	-28.0	-5.54	0.1927	15.1
CMVO (5% stock cap)	19.4	14.6	1.29	-25.9	-4.50	0.0488	6.5

Figures 9 and 10 show how the 5% cap changes the allocation. The main effect is to reduce single-name concentration while keeping the cluster-based diversification structure.

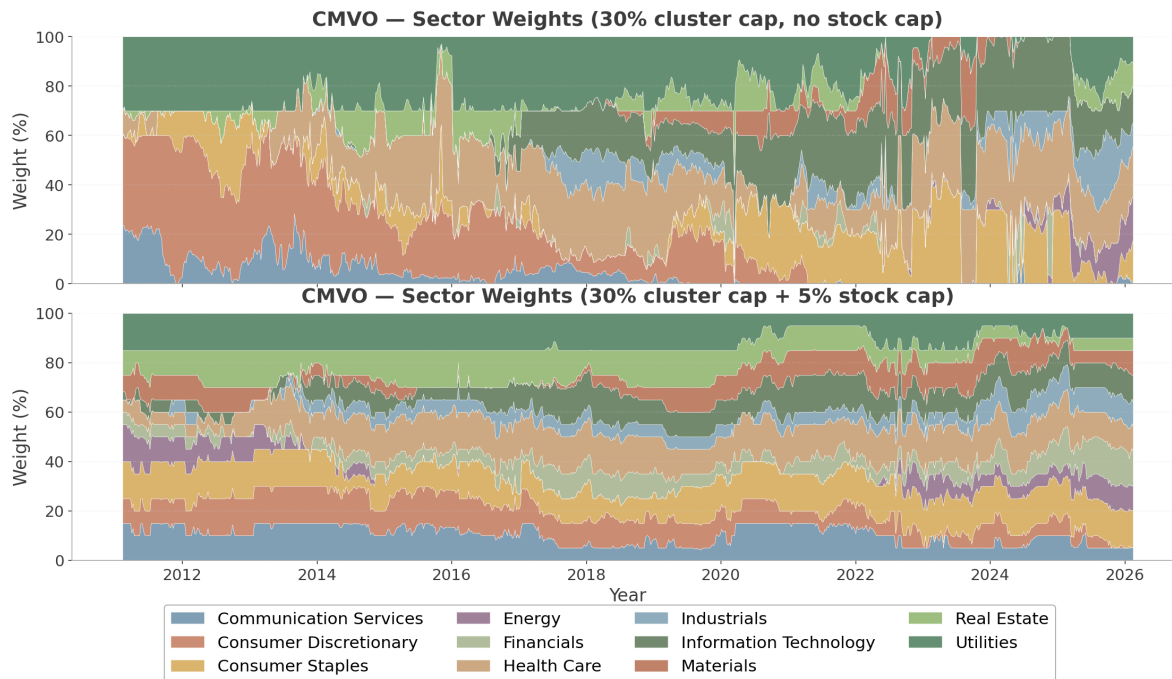


Figure 9: Sector weights for baseline CMVO and CMVO with a 5% stock cap. With a 5% cap the portfolio must hold at least 20 stocks, mechanically spreading weight across all 11 GICS sectors.

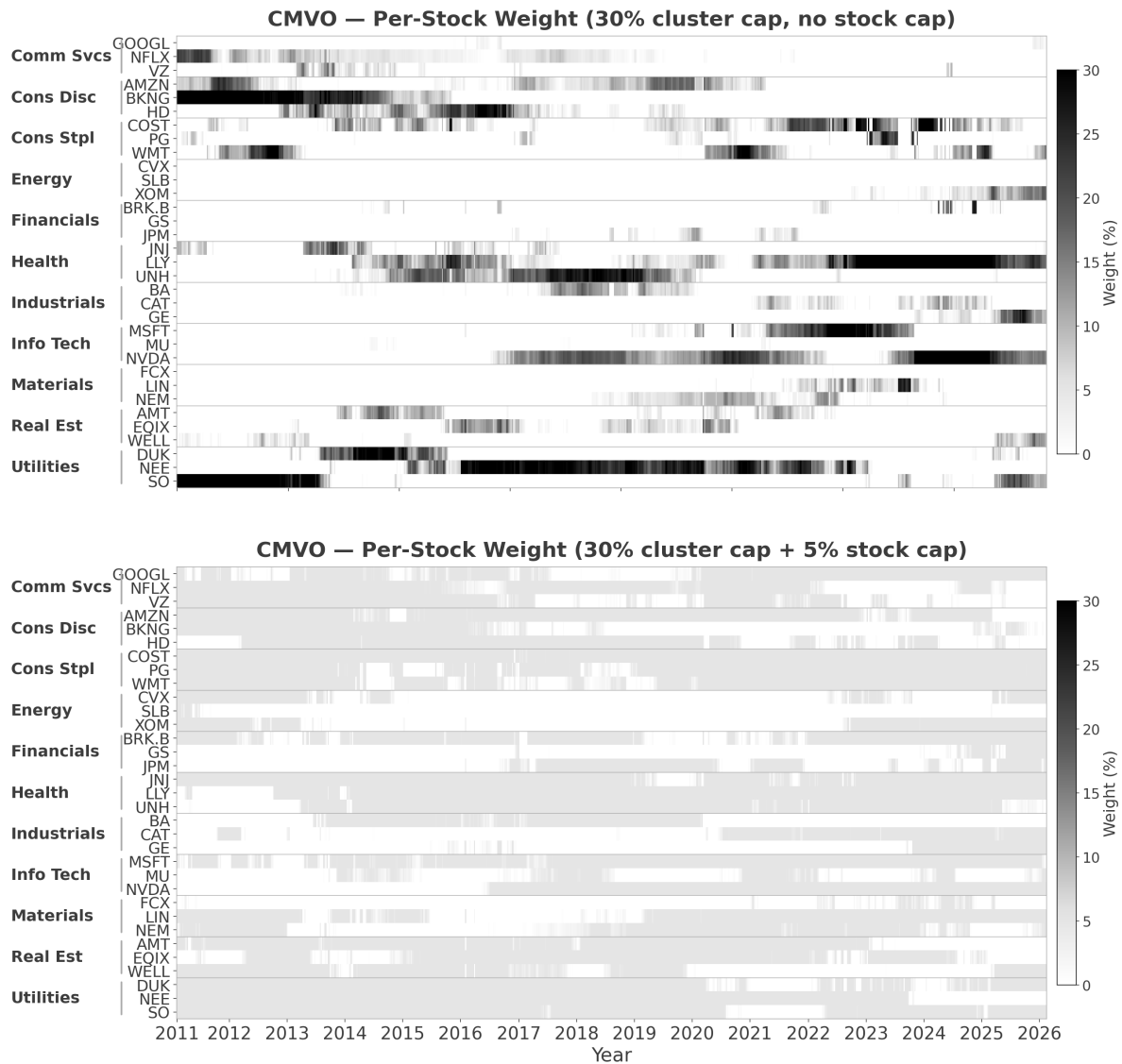


Figure 10: Stock weights for baseline CMVO and CMVO with a 5% stock cap. Both panels use the same color scale so the reduction in concentration is directly visible. Each row within a sector band represents one stock; darker shading indicates higher weight.

4.4 Sensitivity

These sensitivity tests are conducted around the baseline CMVO specification of a 5-year weekly lookback, Ward linkage, $k = 6$ clusters, a 30% cluster cap, sample covariance estimation, and a maximum-Sharpe objective. Table 7 summarizes the main Sharpe ranges.

Table 7: *Sensitivity summary around the baseline CMVO specification. Sharpe ratios remain broadly stable, but stock-level caps have the strongest effect because they directly target residual concentration.*

Hyper-parameter	Range	Sharpe Range	Main Takeaway
Lookback window	2–5 years	1.06–1.16	Moderate sensitivity; five years remains competitive.
Cluster count k	4–9	1.16–1.24	Higher k raises Sharpe but also increases singleton risk.
Cluster cap	20%–100%	1.13–1.18	Tighter caps reduce concentration.
Stock cap	10%–100%	1.16–1.29	Stock-level concentration is the most important practical issue.
Covariance estimator	sample covariance, Ledoit-Wolf shrinkage, oracle-approximating shrinkage [11]	1.16–1.17	Both shrinkage methods are standard covariance-regularization estimators; neither materially changes the result.

The results are robust across a range of parameter choices. The most important sensitivity is the stock-cap dimension, which confirms that residual stock-level concentration is the primary weakness in the baseline specification.

5 Additional Validation

This section adds two robustness checks to the main results: a different validation method, Combinatorial Purged Cross-Validation (CPCV), and a comparison with Nested Clustered Optimization (NCO) in Appendix B.

5.1 Combinatorial Purged Cross-Validation

CPCV⁷ tests robustness by producing a distribution of Sharpe ratios rather than a single backtest number. The CPCV split structure is shown in Figure 11 (Appendix C). Table 8 reports the path-level annualized Sharpe summaries across the 9 stitched out-of-sample paths.

⁷CPCV splits the data into many train/test combinations. A purge removes nearby training observations, and an embargo leaves a short gap after the test window, both to reduce leakage. See [13] for details.

Table 8: *Path-level annualized Sharpe ratios from CPCV across 9 stitched out-of-sample paths. The capped CMVO variant is included because it materially changes the robustness conclusion.*

Strategy	Paths	Mean	Median	Std. Dev.	Min	Max
CMVO	9	0.911	0.903	0.067	0.820	1.027
CMVO (5% stock cap)	9	1.092	1.096	0.024	1.052	1.126
Plain MV	9	0.899	0.876	0.072	0.826	1.045
Equal Weight (1/N)	9	0.999	0.999	0.000	0.999	0.999

The CPCV result is consistent with the main backtest interpretation. CMVO is close to Plain MV, but the 5% stock-cap version is stronger because it directly addresses remaining stock-level concentration. Equal weight remains competitive and very stable.

6 Conclusion

Cluster-based constraints did not produce statistically significant Sharpe gains, but they did reduce concentration and downside risk relative to Plain MV. CMVO consistently delivered lower HHI, shallower maximum drawdowns, and smaller expected shortfall than the unconstrained optimizer.

The results also show that residual stock-level concentration remains an important issue even after cluster constraints are imposed. The additional 5% stock-cap extension improves diversification further and produces the strongest risk-adjusted performance in the sample, although at the cost of some annualized return. More broadly, the findings support the view that practical constraints can improve the real-world behavior of mean-variance portfolios by reducing their tendency toward extreme and unstable positions [7].

Implementation and Verification Note

AI tools (Claude, Codex, and GPT-5.4 Thinking) were used as coding and research assistants throughout this project. To reduce hallucinations, I queried the original reference PDFs through retrieval in Obsidian rather than relying on model memory, and I maintained a local copy of all relevant papers to cross-reference implementations and verify that my understanding was correct. I followed a generate-verify-revise workflow where code outputs, reported numbers, and references were checked against the scripts, the paper sources, and an independent no-context model pass before inclusion. Packages such as vectorbtpro were used to confirm backtest outputs and validate walk-forward splits. All statistical outputs and figures were regenerated locally from the project scripts, and every number in this paper was manually cross-checked against the corresponding CSV output. The full codebase is available in the accompanying submission folder.

A Descriptive Statistics

Table 9: Annualized summary statistics computed from daily simple returns. Mean and standard deviation are annualized; skewness and kurtosis are reported to summarize asymmetry and tail thickness.

Sector	Ticker	Mean %	Std. %	Skew	Kurt.	Min %	Max %
Comm. Services	GOOGL	27.3	30.6	0.60	8.9	-11.6	20.0
	NFLX	40.4	50.5	-0.10	23.0	-40.9	42.2
	VZ	8.5	20.6	0.44	9.3	-8.1	14.6
Cons. Discret.	AMZN	28.9	37.8	0.83	14.6	-21.8	26.9
	HD	16.8	25.8	0.01	10.3	-19.8	14.1
	BKNG	32.4	37.9	1.06	14.3	-17.3	26.1
Consumer Staples	WMT	13.2	20.3	0.28	12.4	-11.4	11.7
	COST	19.1	22.3	-0.06	7.9	-12.4	10.8
	PG	9.4	18.1	0.06	9.8	-8.7	12.0
Energy	XOM	12.3	26.3	0.18	9.9	-13.9	17.2
	CVX	14.0	28.1	0.07	20.5	-22.1	22.7
	SLB	11.9	38.2	-0.23	9.2	-27.4	19.9
Financials	JPM	18.9	35.6	0.94	20.1	-20.7	25.1
	BRK.B	12.2	21.0	0.74	17.2	-10.9	19.3
	GS	18.5	34.5	0.77	18.0	-19.0	26.5
Health Care	LLY	19.4	26.3	0.55	12.1	-14.1	15.7
	UNH	16.3	31.9	0.28	31.2	-22.4	34.8
	JNJ	11.0	17.2	0.18	11.4	-10.0	12.2
Industrials	BA	15.1	34.9	0.26	15.7	-23.9	24.3
	GE	11.1	32.7	0.22	9.0	-15.2	19.7
	CAT	21.5	31.5	0.00	5.4	-14.5	14.7
Info. Technology	NVDA	47.4	48.6	0.20	7.6	-30.7	29.8
	MU	29.8	51.3	0.12	4.4	-19.8	23.4
	MSFT	18.4	26.9	0.23	10.1	-14.7	18.6
Materials	FCX	21.6	51.9	0.15	6.5	-20.3	29.7
	NEM	13.6	38.2	0.32	7.2	-14.7	25.2
	LIN	16.1	24.6	0.31	9.5	-12.6	14.9
Real Estate	AMT	16.6	27.3	0.46	10.0	-15.2	20.2
	WELL	18.1	30.8	0.07	20.6	-24.4	23.2
	EQIX	22.4	34.0	-0.17	16.7	-33.1	17.2
Utilities	NEE	16.8	23.8	0.21	11.2	-13.4	13.9
	DUK	11.9	19.8	0.15	12.9	-11.5	13.3
	SO	11.4	19.2	0.72	22.7	-11.8	18.8

B Nested Clustered Optimization

As an additional comparison, I test Nested Clustered Optimization (NCO), another clustering-based allocation method from Lopez de Prado [14]. Unlike CMVO, which applies cluster constraints to a mean-variance optimizer, NCO uses the hierarchical cluster structure inside the optimization itself rather than imposing an external cluster exposure cap.

The NCO comparison uses the same weekly five-year rolling walk-forward design and the same aligned 18 February 2011 to 13 February 2026 window as the main results. Table 10 compares NCO with CMVO, Plain MV, and equal weight.

Table 10: *Weekly walk-forward comparison including Nested Clustered Optimization. All strategies are aligned to the same 783-week out-of-sample window.*

Strategy	Ann. Ret. %	Ann. Vol. %	Sharpe	Max DD %	ES 5%	HHI	Avg Weekly Turnover %
CMVO	21.6	18.3	1.16	-28.0	-5.54	0.1927	15.1
Plain MV	23.7	20.8	1.13	-31.0	-6.36	0.2650	11.3
NCO	24.6	19.4	1.23	-32.2	-5.93	0.1885	13.7
Equal Weight (1/N)	18.7	16.2	1.14	-32.2	-5.15	0.0303	2.5

NCO has the highest return and Sharpe in this comparison. Diversification is similar to CMVO and better than Plain MV, while CMVO performs better than NCO on drawdown and expected shortfall. Clustering helps, but different clustering methods involve trade-offs: NCO appears to maximize performance, while CMVO offers more controlled downside behavior.

C CPCV Split Structure

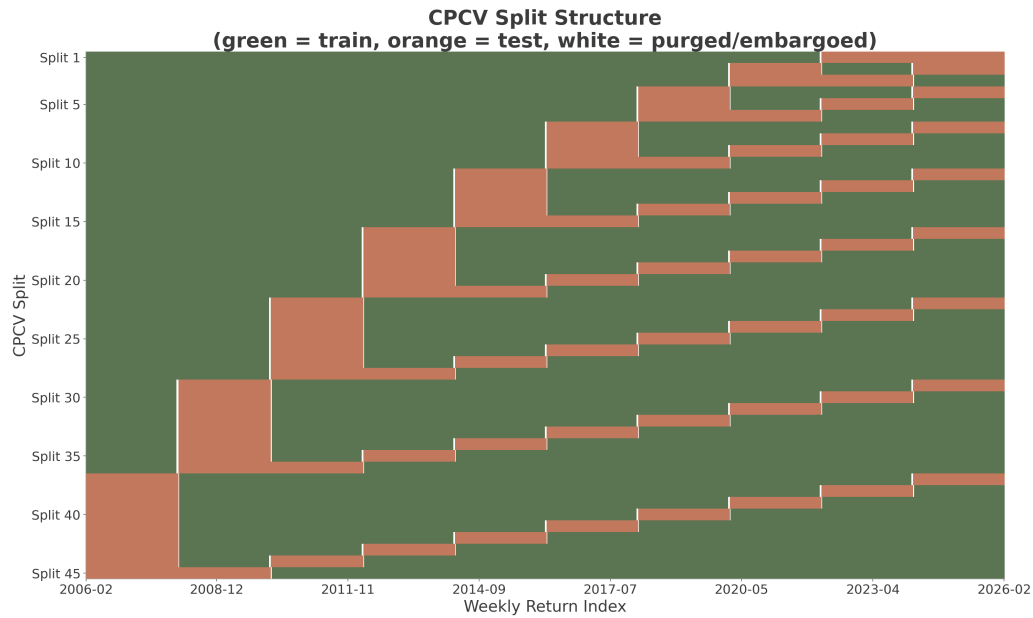


Figure 11: *CPCV split structure with 10 folds, 2 test folds, a 2-week purge, and a 1-week embargo. This design reduces leakage and produces multiple out-of-sample Sharpe paths.*

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